

Distributive Area Model Tiles

Unit 6: Music Studio • Lesson 6-5 • EduWonderLab

UNIT 6 • LESSON 6-5 • TEACHER NOTES

Standard 6.EE.3

FORMAT Area-model tiles + expand/factor

TIME 25–35 min

GROUPING Pairs

TOPIC The Distributive Property

STANDARD 6.EE.3

PREP LEVEL Medium — print tiles and area-model mats

Learning goal *I can use the distributive property to expand and factor expressions.*

Teacher setup

- Print tiles and mats per pair. Students build an area model for each expression.
- They match each model to an expanded and a factored form.
- Play both an expand round and a factor round.

Materials

Paper tiles, Area model mats, Expression cards, Pencil.

Differentiation & language support

- Area model shows the rectangle split into parts (SPED).
- Sentence stem: “I multiplied ___ by each part: ___ and ___.”
- ESOL: label the outside factor and the inside terms.

Key vocabulary (English / Español):

Term	Español	What it means
distribute	distribuir	to multiply across a sum
factor	factorar	to write as a product

Quick teacher check: *Ask a pair to show the area model for Card 3 ($4(2x + 3)$).*

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Name: _____ Date: _____ Class: _____

Your goal: *I can use the distributive property to expand and factor expressions.*

What you need

Paper tiles, Area model mats, Expression cards.

How to play

1. Pick a card. Decide if it is an expand or factor task.
2. Build the area model with tiles.
3. Write the expanded or factored form.
4. Check that both forms are equivalent.

Expand & Factor Cards

To expand, multiply the outside factor by each term. To factor, pull out the greatest common factor.

1. Expand: $3(x + 4)$	2. Expand: $2(5 + a)$
3. Expand: $4(2x + 3)$	4. Expand: $5(x - 2)$
5. Factor: $6x + 9$	6. Factor: $8 + 12x$
7. Factor: $10x + 15$	8. Factor: $14 + 7x$

Record your work (area model + answer):

Explain your thinking

Explain how expanding and factoring are opposite moves.

Sentence frame: *I multiplied ___ by each part: ___ and ___.*

★ **Challenge:** Expand $3(2x + 4)$, then factor your answer back. Do you return to $3(2x + 4)$?

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Expand: $3(x + 4)$ Answer: $3x + 12$	2. Expand: $2(5 + a)$ Answer: $10 + 2a$
3. Expand: $4(2x + 3)$ Answer: $8x + 12$	4. Expand: $5(x - 2)$ Answer: $5x - 10$
5. Factor: $6x + 9$ Answer: $3(2x + 3)$	6. Factor: $8 + 12x$ Answer: $4(2 + 3x)$
7. Factor: $10x + 15$ Answer: $5(2x + 3)$	8. Factor: $14 + 7x$ Answer: $7(2 + x)$

Teaching notes & look-fors

- Expand: multiply outside by each inside term. Factor: divide out the GCF.
- Card 5: GCF of 6 and 9 is 3 $\rightarrow 3(2x + 3)$.